Healthy Skepticism Day 1

The five faults, and the first worked example

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[venue and date]

About Us

Dr. Murray Cantor

PhD, Mathematics — UC Berkeley

Dr. Joel Keenan

MD, Univ of Vermont

- Not anti-vax or MAHA
- From that perspective, there is a lot of confusion about medical claims
- Respect for the challenge of medical research

- Practicing physician with direct patient-care experience
- Translates clinical evidence into plain-language guidance
- Shares a commitment to honest, balanced medical communication

Disclaimer

This presentation is for educational purposes only.

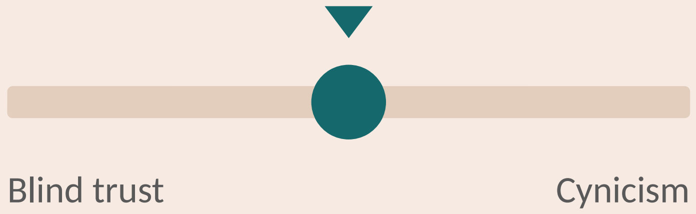
The content does not constitute medical advice and is not a substitute for professional diagnosis, treatment, or care. Nothing presented here should be used to make decisions about your health or the health of others.

Always seek the advice of a qualified physician or other licensed healthcare provider with any questions you may have regarding a medical condition, test result, or treatment. Never disregard professional medical advice or delay seeking it because of something discussed in this presentation.

Why the Course

- Empower patients to:
  - Empower patients to have informed discussions with their doctors
  - Make better health decisions
  - Be aware, but not cynical

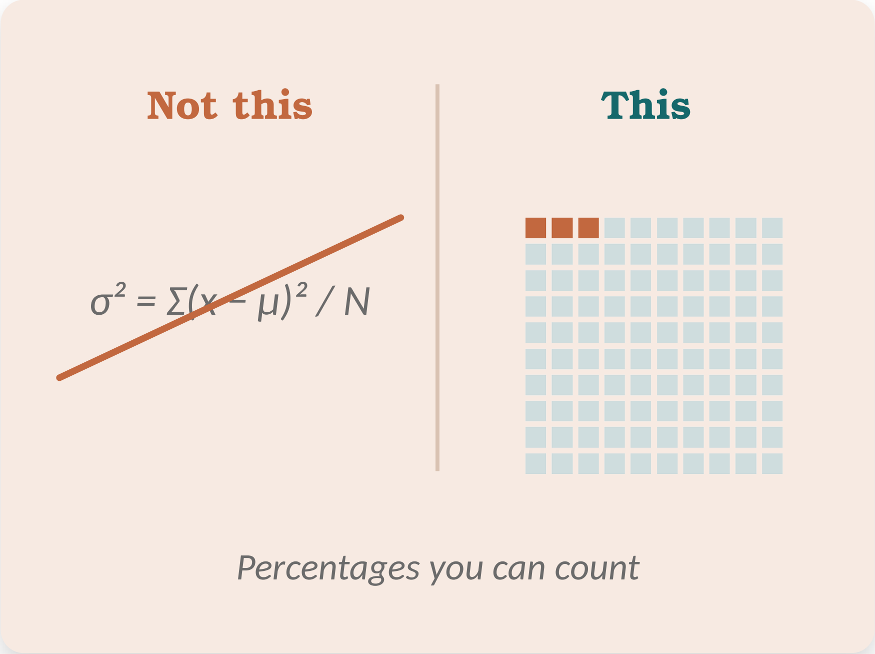
Healthy skepticism



Ask hard questions — without dismissing good science

What to Expect

- We do the math so you don't have to — figures, not formulas
- A lot of real-world examples
- Hopefully, lots of discussion
- All assertions will be backed up
- A companion website with the slides
  - A take-home summary paper for each day
  - Deeper whitepapers on single topics, for the curious



The Five Faults

1 **Relative reporting** Day 1
What does a 36% improvement from taking a drug really mean?

2 **Probability reversal** Day 2
What does a positive COVID test really mean?

3 **Arbitrary categories** Day 3
How concerned should you be if your systolic BP goes from 119 to 120?

4 **The Bernoulli fallacy** Day 3
Why does the published medical advice keep changing?

5 **The causal leap** Day 4
Does bird watching prevent dementia?

Once you see them, you can't unsee them.

Today's Example

1

Relative reporting

A Lipitor advertisement, and what its 36% actually bought

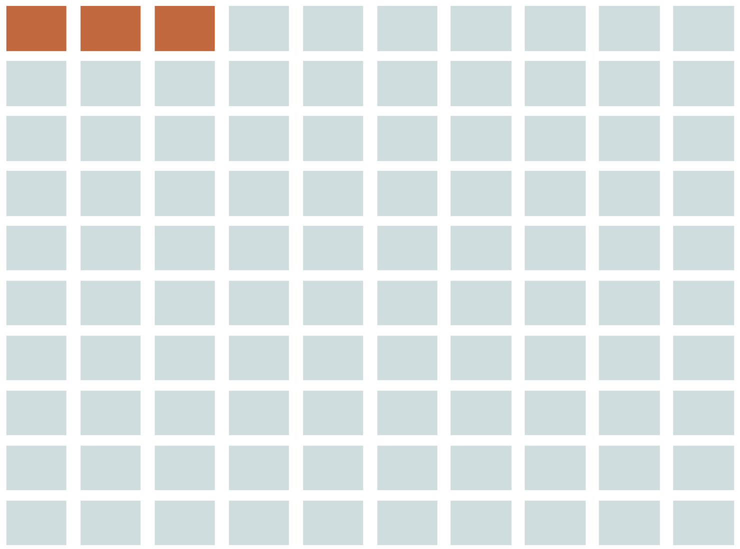
FAULT 1

Relative Reporting

Relative vs. absolute risk

Addressing Uncertainty

- Medicine is uncertain. Every claim is a claim about chances.
- The mathematics of uncertainty is probability theory, which can be abstract and obtuse.
- Fortunately, every concept we need can be explained by counting.
- No equations or formulas. Pictures and a few videos.



Each square is one person. A hundred squares, a hundred people.
Shaded squares are the ones it happens to. Here, 3 in 100.
Illustrative. Every idea in this course is a picture like this.

“Lipitor reduces Your Risk of Heart Attack by 36%!”

Why the number in the ad isn't the number that matters to you

“In patients with multiple risk factors for heart disease, Lipitor reduces the risk of heart attack by 36%.”

36% of what?

A relative cut, not 36 fewer per 100. It shrinks a small base, from 3.0% to 1.9% over about three years.

Your baseline is hidden

The headline never states the starting risk, nor that it applies to high-risk hypertensive patients.

The fine print tells on it

The ad's own asterisk: 3% of placebo patients had a heart attack, versus 2% on Lipitor.

Sources: actual Lipitor advertisement (Pfizer, c. 2003); ASCOT-LLA (Sever et al., Lancet 2003) and Lipitor prescribing label, 36% relative risk reduction, 1.9% vs 3.0%, p = 0.0005.

Next: translate that 36% into the benefit you'd actually get →

Healthy Skepticism

The Same 36%, Shown as 1,000 People

Heart attacks per 1,000 similar patients over about three years, without Lipitor and with it

Without Lipitor

30 in 1,000 have a heart attack

With Lipitor

19 in 1,000 have a heart attack

36% fewer

the relative cut — the headline

11 in 1,000

the absolute benefit · NNT 91

Source: ASCOT-LLA, atorvastatin 10 mg vs placebo, 1.9% vs 3.0% over 3.3 years; absolute reduction 1.1%, NNT 91. Lancet 2003; Lipitor label.

The 36% and the 11 in 1,000 are the same picture, counted two ways.

How to Read the Next “36%” You See

Three questions that turn any relative-risk headline into a real one

“New study: cuts your risk by a third!”

Percent of what?

A percent with no starting number can't be judged. Find the two actual rates being compared.

Find the absolute gap

Subtract them. That difference, per 100 or per 1,000, is the benefit you would actually get.

Then weigh it for you

Against your own baseline risk, the side effects, and the cost. The headline never does this.

The method applies to any relative-risk claim. Worked example throughout: Lipitor, ASCOT-LLA (Lancet 2003).

You now have the tool the headline hoped you wouldn't use →

What we’ve learned

One way a true number can mislead you, and a look ahead.

1 · Relative reporting

The trap: A big-sounding percentage is usually relative — a tiny absolute change dressed up.

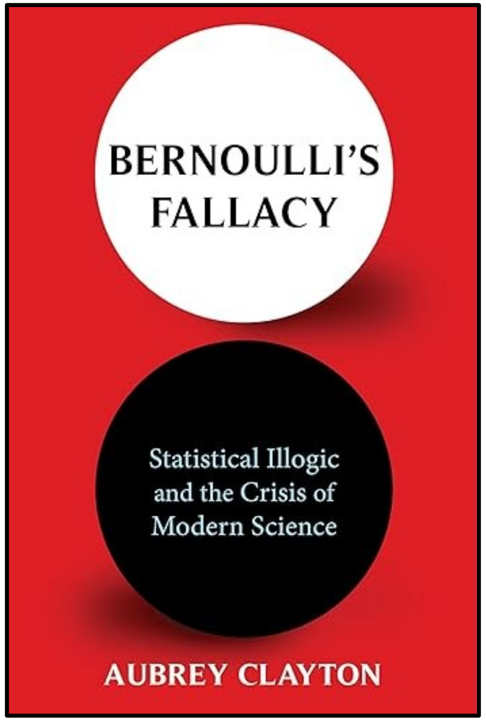
Ask: how many real people does it move? (absolute change, NNT)

2 · Next week: Probability reversal

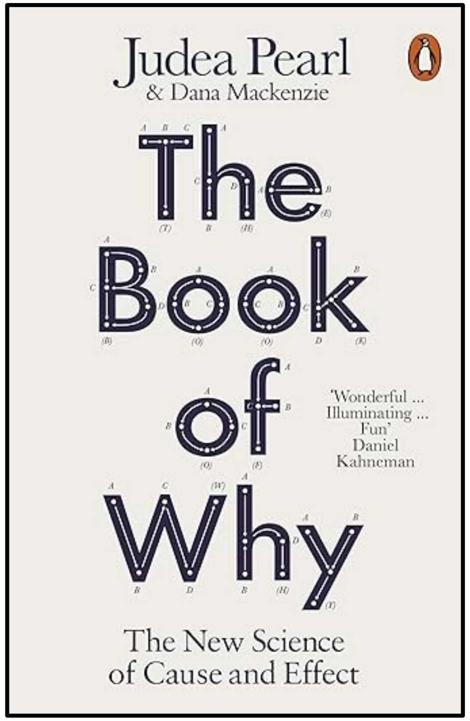
The trap: A number about a crowd gets read as a probability about you.

Ask: given what? What does a positive test actually mean for me?

Good Background Texts



How a 300-year-old logical error fuels today's replication crisis



How to reason about cause and effect

Questions, and a conversation

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Backup Slides

FAULT 3

Arbitrary Categories

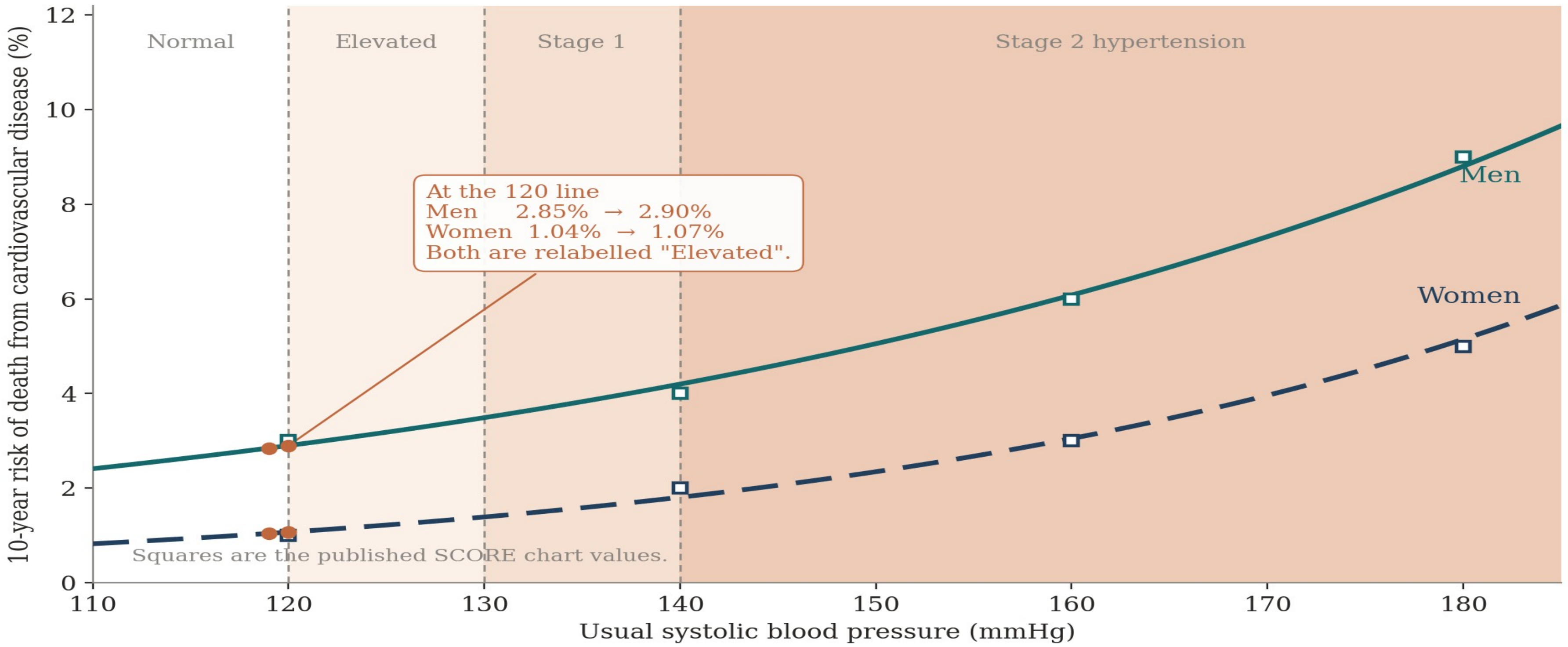
Read the curve, not the cutoff

Where Does “High” Begin?

- Your systolic reads 120. Last year it read 119.
- Nothing about you changed. The label did.
- Every number a doctor measures is continuous. Every label is a box.
- Someone had to decide where the boxes begin, and the curve was not consulted.

Where the categories fall

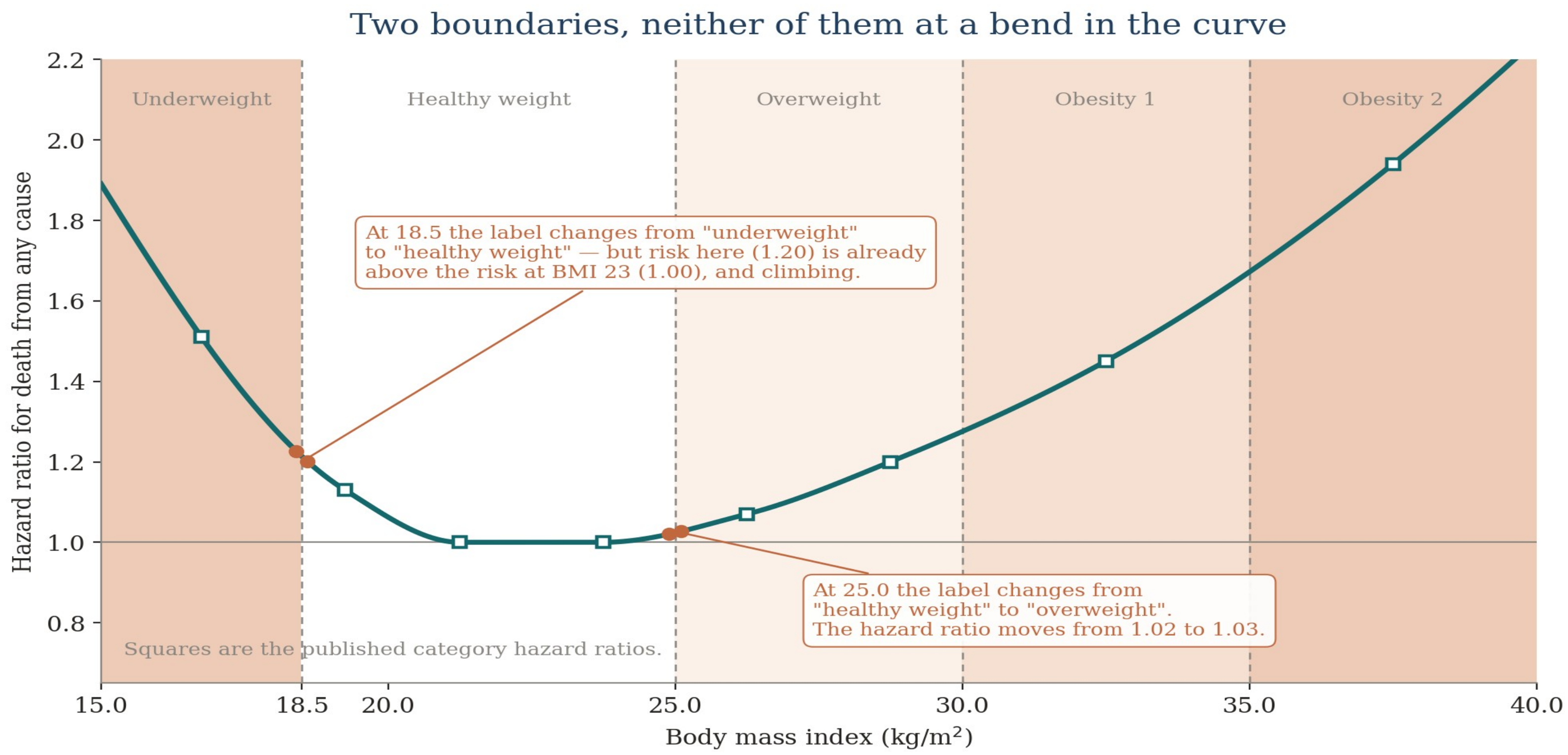
The risk curve does not know where the categories are



One cutoff, two very different absolute risks. The boundary is the same for everyone; what it means is not.

Squares: SCORE low-risk-region chart, age 65, non-smoker, total cholesterol 5 mmol/L (Conroy et al., Eur Heart J 2003; chart ©ESC 2018). Chart cells are whole percentages, so the low end carries up to half a point of rounding, and the steeper female slope is partly an artefact of it. Curves are log-linear fits to each row, consistent with the smooth no-threshold relationship reported by Lewington et al. (Lancet 2002). Category boundaries: ACC/AHA 2017.

The Same Problem in BMI



The curve is flat across the middle and steep at both ends. The category lines fall in neither place.

Squares: Global BMI Mortality Collaboration, Lancet 2016;388:776-786. Hazard ratios relative to BMI 22.5 to under 25, plotted at band midpoints, from 3.95 million never-smokers without chronic disease who survived five years, 385,879 deaths across 189 studies. Curve is a shape-preserving interpolation between those points. Above BMI 25 the slope is steeper in men (1.51 per 5 kg/m^2) than women (1.30). Category boundaries: WHO. The grade 3 band (BMI 40 to 60, HR 2.76) is off the right of this chart.

The Lines Move

- June 1998: the NIH lowered the “overweight” line from 27.8 to 25. About 25 million Americans became overweight overnight, without gaining a pound.
- 2017: the ACC/AHA lowered “hypertension” from 140/90 to 130/80, reclassifying 31 million Americans, 14% of all adults.
- Neither curve moved. Only the lines did.
- Sources: NIH/NHLBI Clinical Guidelines, June 1998 (contemporaneous press reports put the figure near 25 million); Whelton et al., 2017 ACC/AHA guideline, and Muntner et al.

So Why Have Categories At All?

- A box is a decision rule. It tells a busy clinic who to look at twice, and it makes a chart readable in three seconds.
- That is a real use, and a good one. The trouble starts when the box is read as a fact about your body.
- A box cannot tell you how much risk you carry, or how much of it you picked up by crossing the line.
- Ask where you sit on the curve, not which side of the line you are on.

The Same Result, Told Honestly

What a 36% reduction actually buys the person who takes it

"Over about three years, Lipitor prevented a heart attack in roughly 11 of every 1,000 similar patients treated."

The real numbers

30 in 1,000 had a heart attack without it, 19 with it. The drug spared 11 of them.

Number needed to treat

Treat about 91 such patients for three years to prevent one heart attack (NNT 91).

Same fact, two faces

36% and 11 in 1,000 are the identical result. One sounds dramatic, one is honest.

Source: ASCOT-LLA, atorvastatin 10 mg vs placebo, 1.9% vs 3.0% over 3.3 years; absolute reduction 1.1%, NNT 91. Lancet 2003; Lipitor prescribing label.

Next: a real benefit still has to be weighed against your own risk and the harms →

How to Read the Headline

1

Baseline risk

About 3 in 100 untreated patients had a heart attack — a 3.0% absolute risk.

2

What "36%" means

36% OF 3.0% (=1.1%), not 36 points off it — leaving 1.9% risk.

3

Absolute reduction

3.0% – 1.9% = 1.1 percentage points — the real individual benefit.

4

Decision support

Weigh 1.1 points against side effects and cost; higher-risk people gain more.

A "36% risk reduction" only lowers absolute risk from 3.0% to 1.9% — a real gain of 1.1 points, so about 11 of every 1,000 treated actually benefit.

A Real Headline: A Proposed New Flu Shot

The same lesson as the 36% Lipitor ad — now on a real 2026 trial

"New flu shot is 27% more effective for seniors, trial finds"

Illustrative headline, representative of the coverage — not a quotation. The 27% is real.

27% of what?

A relative figure comparing two shots. The base it shrinks is small, so the percentage alone tells you little.

More effective than what?

The yardstick was a standard-dose shot, not the high-dose one adults 65+ are already advised to get.

Your decision needs more

How many people are actually helped, the side effects, and whether any shot beats no shot at all.

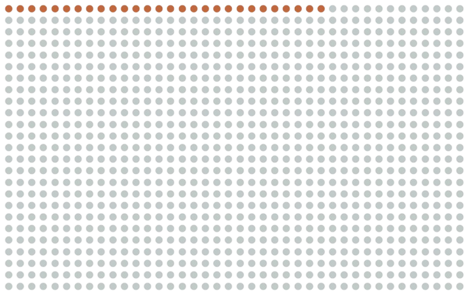
Source: Fluent trial (mRNA-1010), New England Journal of Medicine, May 7, 2026 (NEJMoa2516491).

Next: turn 27% into the number of people it actually means →

From 27% to People

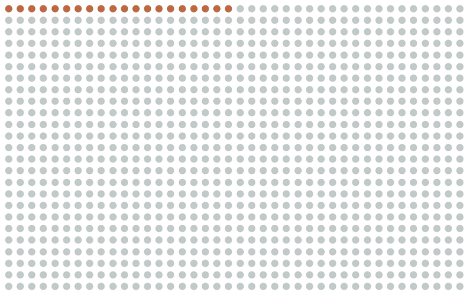
What "27% more effective" looks like per 1,000 seniors vaccinated

Standard-dose shot



28 in 1,000 get flu-like illness

mRNA shot



20 in 1,000 get flu-like illness

8 in 1,000

fewer flu-like illnesses with the mRNA shot — the real, absolute gain.

About 125 seniors switched from standard-dose to mRNA to prevent one extra case (NNV ≈ 125).

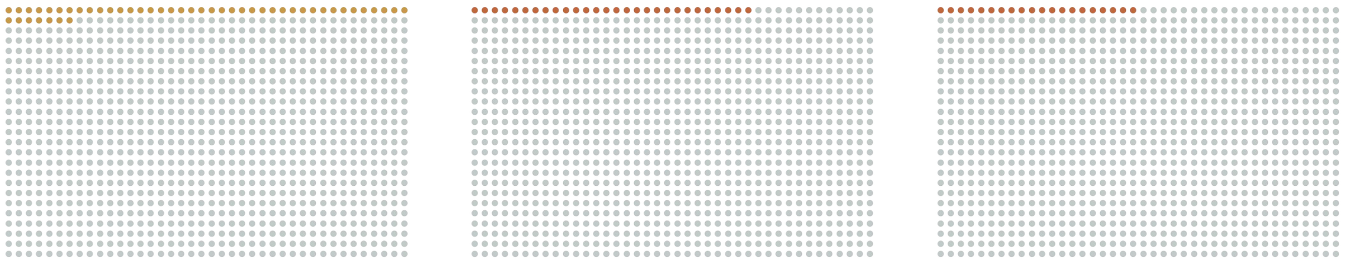
Source: Fluent trial, NEJM 2026 (NEJMoa2516491). Per-protocol flu-like illness 2.0% (mRNA) vs 2.8% (standard-dose).

"27% more effective" = about 8 fewer flu cases per 1,000. Both describe the same result.

The Bigger Question: A Shot, or None?

The headline compares two shots. The bigger choice is a shot versus nothing.

No shot | Standard-dose shot | mRNA shot



~47 in 1,000 (estimated) | 28 in 1,000 | 20 in 1,000

≈19 / 1,000

any shot vs none — the big lever

≈8 / 1,000

mRNA vs standard — the headline

Estimate

no-shot bar has no trial control; from CDC ~40% effectiveness in seniors, and it swings by season

Sources: mRNA vs standard measured in Fluent trial (NEJM 2026). No-shot column modeled from CDC vaccine-effectiveness for ages 65+ (≈40% in 2024-25; 30–41% interim 2025-26).

The big decision is whether to get a shot at all. The headline is only about which one.

Count the Cost, Too

If the benefit gets an absolute number, so should the side effects

Reaction (first week)	mRNA	Standard	Extra
Sore arm	65.8%	29.8%	+36 / 100
Fatigue	45.1%	20.3%	+25 / 100
Headache	37.8%	18.0%	+20 / 100
Muscle aches	35.4%	11.6%	+24 / 100

The honest ledger

Most reactions were mild and lasted a couple of days. Serious events did not differ (2.2% vs 1.9%).

The shot prevents roughly 8 to 27 flu cases per 1,000, while adding several hundred short-lived reactions per 1,000.

Weigh by severity, not by raw count.

Source: Fluent trial, NEJM 2026 — solicited reactions, first week; serious adverse events 2.2% (mRNA) vs 1.9% (standard).

A sore arm is common but brief; a prevented flu case is rarer but heavier. That trade is the decision.

Looking Forward, We will Discuss:

1. The state of medical literature
2. Why public health recommendations often flip
3. The importance of low probabilities
4. Causality versus correlation
5. Reading positive test results (COVID, pregnancy)
6. Responding to measurements outside the normal range
7. ‘Syndrome’, ‘disorder’ and ‘spectrum’: what doctors mean

More Effective Than What?

The comparison behind the headline isn't the shot you'd actually be offered

What the headline implies

A new shot that beats your current flu protection by about a quarter.

What the trial actually tested

The mRNA shot versus a standard-dose shot. Adults 65+ are already advised to get a high-dose or adjuvanted shot, which was not the comparison.

For ages 65 and older, **approval leaned on antibody levels measured against the high-dose shot**, not on a head-to-head count of who actually caught the flu.

Source: Fluent trial, NEJM 2026; FDA Vaccines and Related Biological Products Advisory Committee briefing, June 18, 2026.

A win over the weaker option is not the same as a win over the option you'd actually get.